Electrical Circuit (1)

Introduction (week1 class1)

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FIFTH EDITION

Fundamentals of Electric Circuits



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Systems of Units

TABLE 1.1

Six basic SI units and one derived unit relevant to this text.

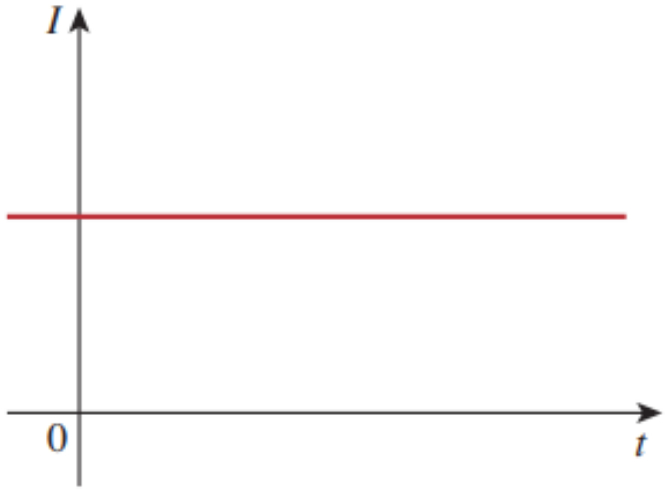
Quantity	Basic unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Luminous intensity	candela	cd
Charge	coulomb	C

TABLE 1.2

The SI prefixes.

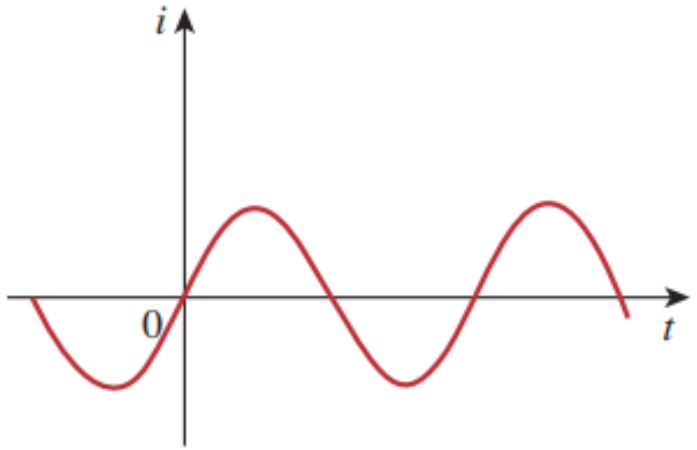
Multiplier	Prefix	Symbol
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{2}	hecto	h
10	deka	da
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a

DC



(a)

AC

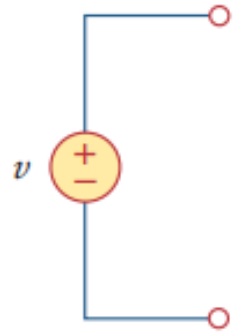


(b)

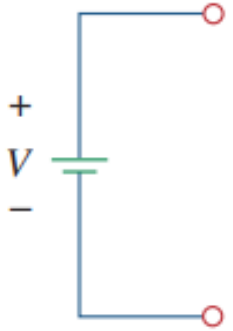
Figure 1.4
Two common types of current: (a) direct current (dc), (b) alternating current (ac).

Circuit Elements

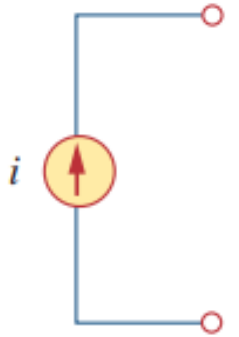
Varying Voltage source



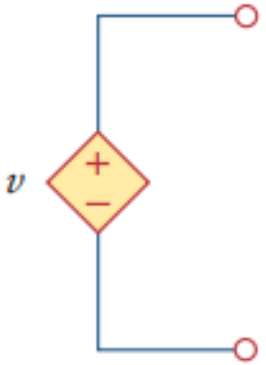
Constant Voltage source



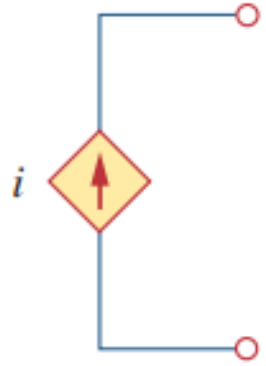
Constant Current source



Dependent Voltage source

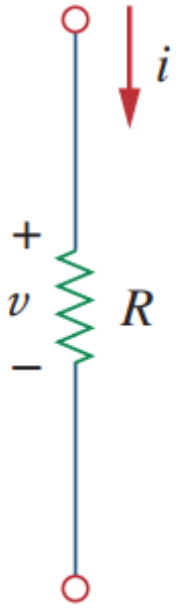


Dependent Current source

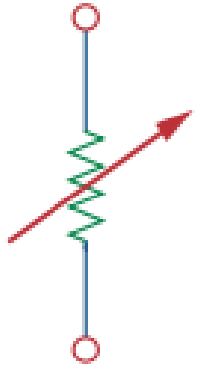


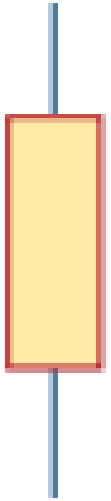
Circuit Elements

Constant Resistance

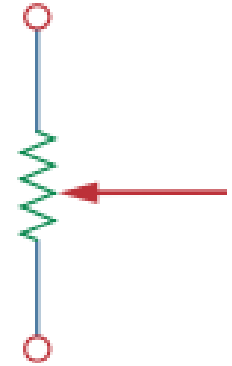


Variable Resistance

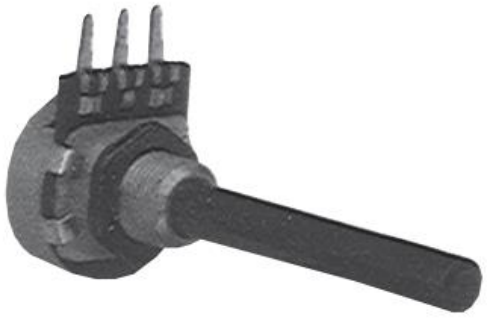




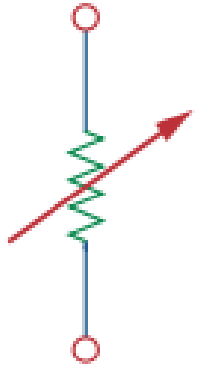
Potentiometer



Circuit Elements

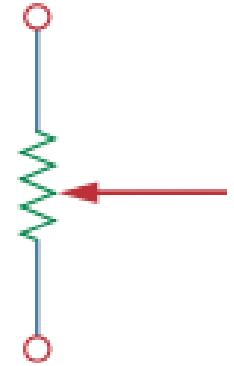


Variable Resistance





Potentiometer



Count the following: -

1. Number of resistance?
2. Number of sources?
3. Number of voltage sources?
4. Number of current sources?
5. Number of dependent sources?
6. Number of independent sources?

